

Intro2SE – Project Proposal – 7

MeoArc

The student team is required to complete the **Project Proposal** documentation for the assigned course project, following the attached template.



Software Engineering Department
Faculty of Information and Technology
VNU-HCMUS

Table of Contents

Objectives	1
1 Member Contribution Assessment	2
2 Preliminary Problem Statement	3
3 Proposed Solution	4
4 Development Plan	5
5 Human Resources & Costing Plan	7
6 Tools setup	8
7 AI Usage Declaration	10
8 Presentation	11
9 Reflective Report	12

Project Proposal

Objectives

This document focus on the following topics:

- ✓ Completing the Project Proposal document with the following sections:
 - Preliminary Problem Statement
 - Proposed Solution
 - Development Plan
 - Human Resources & Costing Plan
- ✓ Understanding the Project Proposal document.
- ✓ This document will be used as input for AI tools to verify the quality of subsequent project artifacts.

All project artifacts must remain consistent and synchronized.

For example, if the project proposal is modified during this phase, a new version of the proposal must be documented.

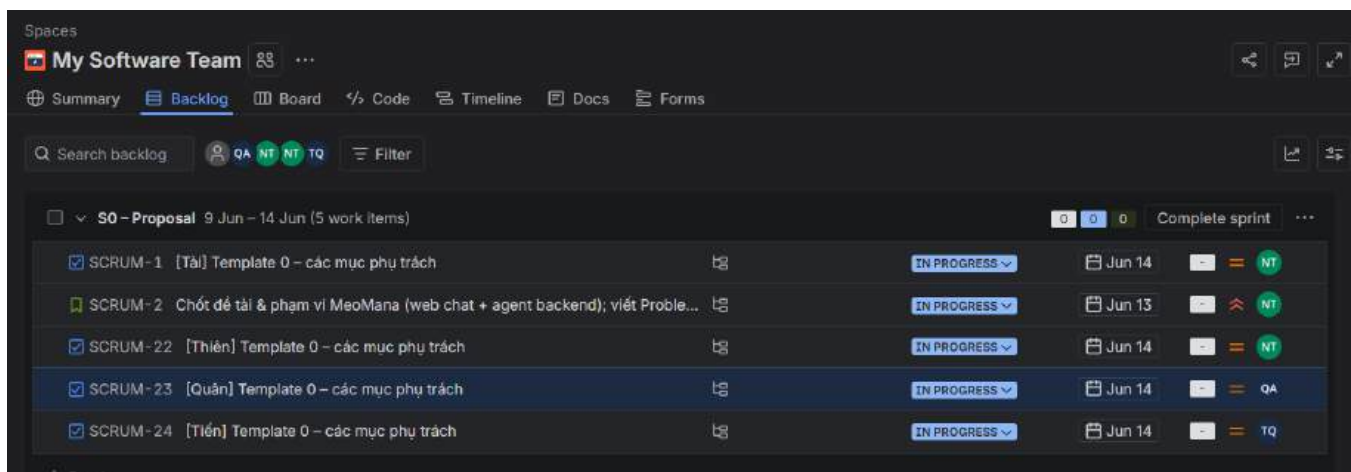
By the conclusion of the project, all versions of every artifact must be submitted to demonstrate the evolution of your work.

1 Member Contribution Assessment

Written by: 24127226 – Phạm Trần Anh Quân

Edited by: 24127250 – Phan Quang Tiến

Reviewed by: 24127529 – Nguyễn Chí Tài



Each task must be assigned to and completed by only one student.

The sum of all student contributions is less than or equal to 100%.

The project tasks includes report writing, self-training, and implementation tasks.

The table below summarizes all Sprint 0 subtasks. Detailed Jira evidence for each member is provided in the screenshots that follow.

Task Name	Assignee	Status	Assigned Date	Completion Date
Dựng GitHub repo theo cấu trúc /src /docs /pa	Phạm Trần Anh Quân	Done	Jun 09, 2026	Jun 14, 2026

Setup Jira board: 5 sprint + deadline	Phạm Trần Anh Quân	Done	Jun 09, 2026	Jun 14, 2026
Tổng hợp Mục 1 – Contribution & evidence	Phạm Trần Anh Quân	Done	Jun 09, 2026	Jun 14, 2026
Kịch bản + ghép + upload video (Mục 8)	Phạm Trần Anh Quân	Done	Jun 09, 2026	Jun 14, 2026
Review Mục 3.2, Mục 6	Phạm Trần Anh Quân	Done	Jun 09, 2026	Jun 14, 2026
Viết Mục 3.2 – Hardware	Phan Quang Tiến	Done	Jun 09, 2026	Jun 14, 2026
Viết Mục 6 – Tools setup	Phan Quang Tiến	Done	Jun 09, 2026	Jun 14, 2026
Viết Mục 7 – AI Usage Declaration	Phan Quang Tiến	Done	Jun 09, 2026	Jun 14, 2026
Viết Mục 9 – Reflective Report	Phan Quang Tiến	Done	Jun 09, 2026	Jun 14, 2026
Review Mục 5	Phan Quang Tiến	Done	Jun 09, 2026	Jun 14, 2026
Quay phần thuyết trình của mình	Phan Quang Tiến	Done	Jun 09, 2026	Jun 14, 2026
Viết Mục 2 – Problem Statement	Nguyễn Chí Tài	Done	Jun 09, 2026	Jun 14, 2026
Viết Mục 4 – Development Plan	Nguyễn Chí Tài	Done	Jun 09, 2026	Jun 14, 2026
Viết Mục 5 – HR & Costing Plan	Nguyễn Chí Tài	Done	Jun 09, 2026	Jun 14, 2026
Nghiên cứu Gmail API + OAuth scopes	Nguyễn Chí Tài	Done	Jun 09, 2026	Jun 14, 2026

Review Mục 1, Mục 8	Nguyễn Chí Tài	Done	Jun 09, 2026	Jun 14, 2026
Quay phần thuyết trình của mình	Nguyễn Chí Tài	Done	Jun 09, 2026	Jun 14, 2026
Viết Mục 3.1.1 – Features (user story mapping)	Nguyễn Ngọc Thiên	Done	Jun 09, 2026	Jun 14, 2026
Viết Mục 3.1.2 – Architecture + vẽ sơ đồ	Nguyễn Ngọc Thiên	Done	Jun 09, 2026	Jun 14, 2026
Self-training: Gemini function calling	Nguyễn Ngọc Thiên	Done	Jun 09, 2026	Jun 14, 2026
Review Mục 2, Mục 7	Nguyễn Ngọc Thiên	Done	Jun 09, 2026	Jun 14, 2026
Quay phần thuyết trình của mình	Nguyễn Ngọc Thiên	Done	Jun 09, 2026	Jun 14, 2026

24127226 – Phạm Trần Anh Quân – 25%

- Task Evidence:

The screenshot displays the Jira interface. On the left, the 'My Software Team' backlog is visible, showing a list of tasks under the 'S0 - Proposal' sprint. The tasks include SCRUM-1 (DONE), SCRUM-2 (IN PROGRESS), SCRUM-22 (IN PROGRESS), SCRUM-23 (DONE), and SCRUM-24 (IN PROGRESS). On the right, the 'Jira work item' details panel is open, showing the following information:

- Assignee:** QA - Quân Phạm Trần Anh
- Parent:** None
- Due date:** Jun 14, 2026
- Labels:** None
- Team:** None
- Start date:** Jun 09, 2026
- Sprint:** S0 - Proposal
- Story point estimate:** None
- Reporter:** QA - Quân Phạm Trần Anh

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' tab with a search bar and filters. The main area lists several sprints: 'S0 - Proposal' (9 Jun - 14 Jun, 5 work items), 'S1 - Requirements' (15 Jun - 28 Jun, 4 work items), 'S2 - Design' (29 Jun - 5 Jul, 4 work items), 'S3 - Testing' (6 Jul - 12 Jul, 5 work items), and 'S4 - Defense Prep' (13 Jul - 20 Jul, 4 work items). The 'S0 - Proposal' sprint is expanded, showing a list of work items: SCRUM-1 (DONE), SCRUM-2 (IN PROGRESS), SCRUM-22 (IN PROGRESS), SCRUM-23 (DONE), and SCRUM-24 (IN PROGRESS). The right sidebar shows the 'Jira work item' detail view for SCRUM-23, titled 'Dựng GitHub repo theo cấu trúc /src /docs /pa'. The details include Assignee (QA - Quân Phạm Trần Anh), Labels (None), Parent (SCRUM-23 [Quản] Template 0 - các mục), Sprint (S0 - Proposal), Story point estimate (None), and Reporter (QA - Quân Phạm Trần Anh).

This screenshot is identical to the one above, showing the Jira interface for the 'My Software Team' space. It displays the backlog with sprints S0 through S4, and the details for work item SCRUM-23. The interface elements, including the sidebar, sprint list, work item list, and the right-hand detail panel, are the same as in the first image.

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' view with a search bar and filters. The main area lists work items for the 'S0 - Proposal' sprint (9 Jun - 14 Jun, 5 work items). The items are:

- SCRUM-1 [Tải] Template 0 - các mục p... (DONE, Jun 14, NT)
- SCRUM-2 Chốt đề tài & phạm vi MeoMa... (IN PROGRESS, Jun 13, NT)
- SCRUM-22 [Thiên] Template 0 - các mục... (IN PROGRESS, Jun 14, NT)
- SCRUM-23 [Quản] Template 0 - các mục... (DONE, Jun 14, QA)
- SCRUM-24 [Tiến] Template 0 - các mục... (IN PROGRESS, Jun 14, TQ)

The right sidebar shows the 'Jira work item' detail view for 'Tổng hợp Mục 1 - Contribution & evidence'. The details include:

- Assignee: QA - Quân Phạm Trần Anh
- Labels: None
- Parent: SCRUM-23 [Quản] Template 0 - các mục
- Sprint: S0 - Proposal
- Story point estimate: None
- Reporter: QA - Quân Phạm Trần Anh

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' view with a search bar and filters. The main area lists work items for the 'S0 - Proposal' sprint (9 Jun - 14 Jun, 5 work items). The items are:

- SCRUM-1 [Tải] Template 0 - các mục p... (DONE, Jun 14, NT)
- SCRUM-2 Chốt đề tài & phạm vi MeoMa... (IN PROGRESS, Jun 13, NT)
- SCRUM-22 [Thiên] Template 0 - các mục... (IN PROGRESS, Jun 14, NT)
- SCRUM-23 [Quản] Template 0 - các mục... (DONE, Jun 14, QA)
- SCRUM-24 [Tiến] Template 0 - các mục... (IN PROGRESS, Jun 14, TQ)

The right sidebar shows the 'Jira work item' detail view for 'Kịch bản + ghép + upload video (Mục 8)'. The details include:

- Assignee: QA - Quân Phạm Trần Anh
- Labels: None
- Parent: SCRUM-23 [Quản] Template 0 - các mục
- Sprint: S0 - Proposal
- Story point estimate: None
- Reporter: QA - Quân Phạm Trần Anh

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays the Backlog with a search bar and filters. The main area shows a list of work items under the 'S0 - Proposal' sprint. The right sidebar shows the details of a selected work item, 'Review Mục 3.2, Mục 6'.

Work Item ID	Summary	Status	Due Date	Assignee
SCRUM-1	[Tải] Template 0 - các mục p...	DONE	Jun 14	NT
SCRUM-2	Chốt đề tài & phạm vi MeoMa...	IN PROGRESS	Jun 13	NT
SCRUM-22	[Thiên] Template 0 - các mục...	IN PROGRESS	Jun 14	NT
SCRUM-23	[Quản] Template 0 - các mục...	DONE	Jun 14	QA
SCRUM-24	[Tiến] Template 0 - các mục...	IN PROGRESS	Jun 14	TQ

Jira Work Item Details: Review Mục 3.2, Mục 6

- Assignee:** QA - Quản Phạm Trần Anh
- Labels:** None
- Parent:** SCRUM-23 [Quản] Template 0 - các mục
- Sprint:** S0 - Proposal
- Story point estimate:** None
- Reporter:** QA - Quản Phạm Trần Anh

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays the Backlog with a search bar and filters. The main area shows a list of work items under the 'S0 - Proposal' sprint. The right sidebar shows the details of a selected work item, 'Quay phần thuyết trình của mình'.

Work Item ID	Summary	Status	Due Date	Assignee
SCRUM-1	[Tải] Template 0 - các mục p...	DONE	Jun 14	NT
SCRUM-2	Chốt đề tài & phạm vi MeoMa...	IN PROGRESS	Jun 13	NT
SCRUM-22	[Thiên] Template 0 - các mục...	IN PROGRESS	Jun 14	NT
SCRUM-23	[Quản] Template 0 - các mục...	DONE	Jun 14	QA
SCRUM-24	[Tiến] Template 0 - các mục...	IN PROGRESS	Jun 14	TQ

Jira Work Item Details: Quay phần thuyết trình của mình

- Assignee:** QA - Quản Phạm Trần Anh
- Labels:** None
- Parent:** SCRUM-23 [Quản] Template 0 - các mục
- Sprint:** S0 - Proposal
- Story point estimate:** None
- Reporter:** QA - Quản Phạm Trần Anh

24127250 - Phan Quang Tiến - 25%

- Task Evidence:

Spaces
My Software Team

Summary Backlog Board Code Timeline Docs Forms

Search backlog Filter

S0 - Proposal 9 Jun - 14 Jun (5 work items) Complete sprint

Item	Status	Due Date	Assignee
SCRUM-1 [Tài] Template 0 - các mục p...	DONE	Jun 14	NT
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	DONE	Jun 13	NT
SCRUM-22 [Thiên] Template 0 - các mục...	DONE	Jun 14	NT
SCRUM-23 [Quân] Template 0 - các mục...	DONE	Jun 14	QA
SCRUM-24 [Tiến] Template 0 - các mục...	DONE	Jun 14	TQ

+ Create

S1 - Requirements 15 Jun - 28 Jun (4 work items) Start sprint

S2 - Design 29 Jun - 5 Jul (4 work items) Start sprint

S3 - Testing 6 Jul - 12 Jul (5 work items) Start sprint

S4 - Defense Prep 13 Jul - 20 Jul (4 work items) Start sprint

Jira work item

Your pinned fields

Details

Assignee: TQ Tiến Phan Quang
Assign to me

Parent: None

Due date: Jun 14, 2026

Labels: None

Team: None

Start date: Jun 09, 2026

Sprint: S0 - Proposal

Story point estimate: None

Reporter: QA Quân Phạm Trần Anh

Development

Spaces
My Software Team

Summary Backlog Board Code Timeline Docs Forms

Search backlog Filter

S0 - Proposal 9 Jun - 14 Jun (5 work items) Complete sprint

Item	Status	Due Date	Assignee
SCRUM-1 [Tài] Template 0 - các mục p...	DONE	Jun 14	NT
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	DONE	Jun 13	NT
SCRUM-22 [Thiên] Template 0 - các mục...	DONE	Jun 14	NT
SCRUM-23 [Quân] Template 0 - các mục...	DONE	Jun 14	QA
SCRUM-24 [Tiến] Template 0 - các mục...	DONE	Jun 14	TQ

+ Create

S1 - Requirements 15 Jun - 28 Jun (4 work items) Start sprint

S2 - Design 29 Jun - 5 Jul (4 work items) Start sprint

S3 - Testing 6 Jul - 12 Jul (5 work items) Start sprint

S4 - Defense Prep 13 Jul - 20 Jul (4 work items) Start sprint

Jira work item

Viết Mục 3.2 - Hardware

+ Done Agents Done Improve Subtask

Description
Add a description...

Linked work items
Add linked work item

Details

Assignee: TQ Tiến Phan Quang
Assign to me

Labels: None

Parent: SCRUM-24 [Tiến] Template 0 - các mục

Sprint: S0 - Proposal

Story point estimate: None

Reporter: QA Quân Phạm Trần Anh

The screenshot shows the Jira interface for the 'My Software Team' space. The left sidebar displays the backlog with a search bar and filters. The main area shows a list of work items under the 'S0 - Proposal' sprint. The right sidebar shows the details for the selected work item 'Viết Mục 6 – Tools setup'.

Backlog:

- S0 - Proposal** (9 Jun - 14 Jun, 5 work items): Complete sprint
 - SCRUM-1 [Tai] Template 0 - các mục p... (DONE, Jun 14, NT)
 - SCRUM-2 Chốt đề tài & phạm vi MeoMa... (DONE, Jun 13, NT)
 - SCRUM-22 [Thiên] Template 0 - các mục... (DONE, Jun 14, NT)
 - SCRUM-23 [Quân] Template 0 - các mục... (DONE, Jun 14, QA)
 - SCRUM-24 [Tiến] Template 0 - các mục... (DONE, Jun 14, TQ)
- S1 - Requirements** (15 Jun - 28 Jun, 4 work items): Start sprint
- S2 - Design** (29 Jun - 5 Jul, 4 work items): Start sprint
- S3 - Testing** (6 Jul - 12 Jul, 5 work items): Start sprint
- S4 - Defense Prep** (13 Jul - 20 Jul, 4 work items): Start sprint

Work Item Details: Viết Mục 6 – Tools setup

- Description:** Add a description...
- Linked work items:** Add linked work item
- Details:**
 - Assignee: TQ - Tiến Phan Quang (Assign to me)
 - Labels: None
 - Parent: SCRUM-24 [Tiến] Template 0 - các mục...
 - Sprint: S0 - Proposal
 - Story point estimate: None
 - Reporter: QA - Quân Phạm Trần Anh

The screenshot shows the Jira interface for the 'My Software Team' space. The left sidebar displays the backlog with a search bar and filters. The main area shows a list of work items under the 'S0 - Proposal' sprint. The right sidebar shows the details for the selected work item 'Viết Mục 7 – AI Usage Declaration'.

Backlog:

- S0 - Proposal** (9 Jun - 14 Jun, 5 work items): Complete sprint
 - SCRUM-1 [Tai] Template 0 - các mục p... (DONE, Jun 14, NT)
 - SCRUM-2 Chốt đề tài & phạm vi MeoMa... (DONE, Jun 13, NT)
 - SCRUM-22 [Thiên] Template 0 - các mục... (DONE, Jun 14, NT)
 - SCRUM-23 [Quân] Template 0 - các mục... (DONE, Jun 14, QA)
 - SCRUM-24 [Tiến] Template 0 - các mục... (DONE, Jun 14, TQ)
- S1 - Requirements** (15 Jun - 28 Jun, 4 work items): Start sprint
- S2 - Design** (29 Jun - 5 Jul, 4 work items): Start sprint
- S3 - Testing** (6 Jul - 12 Jul, 5 work items): Start sprint
- S4 - Defense Prep** (13 Jul - 20 Jul, 4 work items): Start sprint

Work Item Details: Viết Mục 7 – AI Usage Declaration

- Description:** Add a description...
- Linked work items:** Add linked work item
- Details:**
 - Assignee: TQ - Tiến Phan Quang (Assign to me)
 - Labels: None
 - Parent: SCRUM-24 [Tiến] Template 0 - các mục...
 - Sprint: S0 - Proposal
 - Story point estimate: None
 - Reporter: QA - Quân Phạm Trần Anh

Spaces

My Software Team

Summary Backlog Board Code Timeline Docs Forms

Search backlog QA NT NT TQ Filter

S0 - Proposal 9 Jun - 14 Jun (5 work items) 0 0 0 Complete sprint ...

Item	Status	Due Date	Assignee
SCRUM-1 [Tài] Template 0 - các mục p...	DONE	Jun 14	NT
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	DONE	Jun 13	NT
SCRUM-22 [Thiên] Template 0 - các mục...	DONE	Jun 14	NT
SCRUM-23 [Quân] Template 0 - các mục...	DONE	Jun 14	QA
SCRUM-24 [Tiến] Template 0 - các mục...	DONE	Jun 14	TQ

+ Create

S1 - Requirements 15 Jun - 28 Jun (4 work items) 0 0 0 Start sprint ...

S2 - Design 29 Jun - 5 Jul (4 work items) 0 0 0 Start sprint ...

S3 - Testing 6 Jul - 12 Jul (5 work items) 0 0 0 Start sprint ...

S4 - Defense Prep 13 Jul - 20 Jul (4 work items) 0 0 0 Start sprint ...

Jira work item

Viết Mục 9 - Reflective Report

+ Done Agents Done Improve Subtask

Description
Add a description...

Linked work items
Add linked work item

Details

Assignee	TQ Tiến Phan Quang
Labels	None
Parent	SCRUM-24 [Tiến] Template 0 - các mục
Sprint	S0 - Proposal
Story point estimate	None
Reporter	QA Quân Phạm Trần Anh

Spaces

My Software Team

Summary Backlog Board Code Timeline Docs Forms

Search backlog QA NT NT TQ Filter

S0 - Proposal 9 Jun - 14 Jun (5 work items) 0 0 0 Complete sprint ...

Item	Status	Due Date	Assignee
SCRUM-1 [Tài] Template 0 - các mục p...	DONE	Jun 14	NT
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	DONE	Jun 13	NT
SCRUM-22 [Thiên] Template 0 - các mục...	DONE	Jun 14	NT
SCRUM-23 [Quân] Template 0 - các mục...	DONE	Jun 14	QA
SCRUM-24 [Tiến] Template 0 - các mục...	DONE	Jun 14	TQ

+ Create

S1 - Requirements 15 Jun - 28 Jun (4 work items) 0 0 0 Start sprint ...

S2 - Design 29 Jun - 5 Jul (4 work items) 0 0 0 Start sprint ...

S3 - Testing 6 Jul - 12 Jul (5 work items) 0 0 0 Start sprint ...

S4 - Defense Prep 13 Jul - 20 Jul (4 work items) 0 0 0 Start sprint ...

Jira work item

Review Mục 5

+ Done Agents Done Improve Subtask

Description
Add a description...

Linked work items
Add linked work item

Details

Assignee	TQ Tiến Phan Quang
Labels	None
Parent	SCRUM-24 [Tiến] Template 0 - các mục
Sprint	S0 - Proposal
Story point estimate	None
Reporter	QA Quân Phạm Trần Anh

The screenshot displays the Jira Software interface for a team named "My Software Team". The left sidebar shows the "Backlog" view with a search bar and a filter. The main area shows a backlog of work items for the "S0 - Proposal" sprint, which is currently in progress. The work items are listed with their status (DONE, IN PROGRESS, or NOT STARTED) and assigned to team members (NT, TQ, QA). The right sidebar shows the "Jira work item" detail view for the selected item, "Quay phần thuyết trình của mình". The detail view includes fields for Assignee (TQ - Tiến Phan Quang), Labels (None), Parent (SCRUM-24 [Tiến] Template 0 - các mục...), Sprint (S0 - Proposal), Story point estimate (None), and Reporter (QA - Quân Phạm Trần Anh).

24127529 - Nguyễn Chí Tài - 25%

- Task Evidence:

The screenshot displays the Jira Software interface for a team named "My Software Team". The left sidebar shows the "Backlog" view with a search bar and a filter. The main area shows a backlog of work items for the "S0 - Proposal" sprint, which is currently in progress. The work items are listed with their status (DONE, IN PROGRESS, or NOT STARTED) and assigned to team members (NT, TQ, QA). The right sidebar shows the "Jira work item" detail view for the selected item, "Quay phần thuyết trình của mình". The detail view includes fields for Assignee (NT - Nguyễn Chí Tài), Labels (None), Parent (None), Due date (Jun 14, 2026), Labels (None), Team (None), Start date (Jun 09, 2026), Sprint (S0 - Proposal), Story point estimate (None), and Reporter (TQ - Tiến Phan Quang).

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' view with a search bar and a filter. The main content area lists several work items under the 'S0 - Proposal' sprint, including SCRUM-1 (DONE), SCRUM-2 (IN PROGRESS), SCRUM-22 (IN PROGRESS), SCRUM-23 (DONE), and SCRUM-24 (IN PROGRESS). The right sidebar shows the details for the selected work item 'Viết Mục 2 - Problem Statement'. The details include the assignee 'NT Nguyễn Chí Tài', labels 'None', parent 'SCRUM-1 [Tài] Template 0 - các mục phụ', sprint 'S0 - Proposal', story point estimate 'None', and reporter 'QA Quân Phạm Trần Anh'.

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' view with a search bar and a filter. The main content area lists several work items under the 'S0 - Proposal' sprint, including SCRUM-1 (DONE), SCRUM-2 (IN PROGRESS), SCRUM-22 (IN PROGRESS), SCRUM-23 (DONE), and SCRUM-24 (IN PROGRESS). The right sidebar shows the details for the selected work item 'Viết Mục 4 - Development Plan'. The details include the assignee 'NT Nguyễn Chí Tài', labels 'None', parent 'SCRUM-1 [Tài] Template 0 - các mục phụ', sprint 'S0 - Proposal', story point estimate 'None', and reporter 'QA Quân Phạm Trần Anh'.

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' tab with a search bar and filters. The main area lists several sprints: 'S0 - Proposal' (9 Jun - 14 Jun, 5 work items), 'S1 - Requirements' (15 Jun - 28 Jun, 4 work items), 'S2 - Design' (29 Jun - 5 Jul, 4 work items), 'S3 - Testing' (6 Jul - 12 Jul, 5 work items), and 'S4 - Defense Prep' (13 Jul - 20 Jul, 4 work items). The 'S0 - Proposal' sprint is expanded, showing a list of work items with their status (DONE, IN PROGRESS) and assignees (NT, QA, TQ).

The right sidebar shows the 'Jira work item' details for the item 'Viết Mục 5 - HR & Costing Plan'. The details include:

- Description:** Add a description...
- Linked work items:** Add linked work item
- Details:**
 - Assignee:** NT Nguyễn Chí Tài (Assign to me)
 - Labels:** None
 - Parent:** SCRUM-1 [Tài] Template 0 - các mục phụ
 - Sprint:** S0 - Proposal
 - Story point estimate:** None
 - Reporter:** QA Quân Phạm Trần Anh

The screenshot displays the Jira interface for the 'My Software Team' space, similar to the one above. The left sidebar shows the 'Backlog' tab with a search bar and filters. The main area lists several sprints: 'S0 - Proposal' (9 Jun - 14 Jun, 5 work items), 'S1 - Requirements' (15 Jun - 28 Jun, 4 work items), 'S2 - Design' (29 Jun - 5 Jul, 4 work items), 'S3 - Testing' (6 Jul - 12 Jul, 5 work items), and 'S4 - Defense Prep' (13 Jul - 20 Jul, 4 work items). The 'S0 - Proposal' sprint is expanded, showing a list of work items with their status (DONE, IN PROGRESS) and assignees (NT, QA, TQ).

The right sidebar shows the 'Jira work item' details for the item 'Nghiên cứu Gmail API + OAuth scopes'. The details include:

- Description:** Add a description...
- Linked work items:** Add linked work item
- Details:**
 - Assignee:** NT Nguyễn Chí Tài (Assign to me)
 - Labels:** None
 - Parent:** SCRUM-1 [Tài] Template 0 - các mục phụ
 - Sprint:** S0 - Proposal
 - Story point estimate:** None
 - Reporter:** QA Quân Phạm Trần Anh

Spaces

My Software Team

Summary Backlog Board Code Timeline Docs Forms

Search backlog QA NT NT TQ Filter

S0 - Proposal 9 Jun - 14 Jun (5 work items) 0 0 0 Complete sprint

Item	Status	Due Date	Assignee
SCRUM-1 [Tài] Template 0 - các mục p...	DONE	Jun 14	NT
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	IN PROGRESS	Jun 13	NT
SCRUM-22 [Thiên] Template 0 - các mục...	IN PROGRESS	Jun 14	NT
SCRUM-23 [Quân] Template 0 - các mục...	DONE	Jun 14	QA
SCRUM-24 [Tiến] Template 0 - các mục...	IN PROGRESS	Jun 14	TQ

+ Create

S1 - Requirements 15 Jun - 28 Jun (4 work items) 0 0 0 Start sprint

S2 - Design 29 Jun - 5 Jul (4 work items) 0 0 0 Start sprint

S3 - Testing 6 Jul - 12 Jul (5 work items) 0 0 0 Start sprint

S4 - Defense Prep 13 Jul - 20 Jul (4 work items) 0 0 0 Start sprint

Jira work item

Review Mục 1, Mục 8

Done Agents Done Improve Subtask

Description
Add a description...

Linked work items
Add linked work item

Details

Assignee: NT Nguyễn Chí Tài
Assign to me

Labels: None

Parent: SCRUM-1 [Tài] Template 0 - các mục phụ

Sprint: S0 - Proposal

Story point estimate: None

Reporter: QA Quân Phạm Trần Anh

Spaces

My Software Team

Summary Backlog Board Code Timeline Docs Forms

Search backlog QA NT NT TQ Filter

S0 - Proposal 9 Jun - 14 Jun (5 work items) 0 0 0 Complete sprint

Item	Status	Due Date	Assignee
SCRUM-1 [Tài] Template 0 - các mục p...	DONE	Jun 14	NT
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	IN PROGRESS	Jun 13	NT
SCRUM-22 [Thiên] Template 0 - các mục...	IN PROGRESS	Jun 14	NT
SCRUM-23 [Quân] Template 0 - các mục...	DONE	Jun 14	QA
SCRUM-24 [Tiến] Template 0 - các mục...	IN PROGRESS	Jun 14	TQ

+ Create

S1 - Requirements 15 Jun - 28 Jun (4 work items) 0 0 0 Start sprint

S2 - Design 29 Jun - 5 Jul (4 work items) 0 0 0 Start sprint

S3 - Testing 6 Jul - 12 Jul (5 work items) 0 0 0 Start sprint

S4 - Defense Prep 13 Jul - 20 Jul (4 work items) 0 0 0 Start sprint

Jira work item

Quay phần thuyết trình của mình

Done Agents Done Improve Subtask

Description
Add a description...

Linked work items
Add linked work item

Details

Assignee: NT Nguyễn Chí Tài
Assign to me

Labels: None

Parent: SCRUM-1 [Tài] Template 0 - các mục phụ

Sprint: S0 - Proposal

Story point estimate: None

Reporter: QA Quân Phạm Trần Anh

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays a backlog with several items, including 'S0 - Proposal' (9 Jun - 14 Jun, 5 work items) and 'S1 - Requirements' (15 Jun - 28 Jun, 4 work items). The main panel shows the 'S0 - Proposal' sprint with a list of tasks. The right panel shows the details of a Jira work item titled 'Chốt đề tài & phạm vi MeoArc (web chat + agent backend); viết Problem Statement (Mục 2); quay video thuyết trình; viết Reflective Report'. The work item is assigned to 'NT' (Nguyễn Chí Tài) and has a status of 'DONE'.

The screenshot shows the Jira interface for 'My Software Team'. The left sidebar displays a backlog with several items, including 'S0 - Proposal' (9 Jun - 14 Jun, 5 work items) and 'S1 - Requirements' (15 Jun - 28 Jun, 4 work items). The main panel shows the 'S0 - Proposal' sprint with a list of tasks. The right panel shows the details of a Jira work item titled 'Chốt đề tài & phạm vi MeoArc (web chat + agent backend); viết Problem Statement (Mục 2); quay video thuyết trình; viết Reflective Report'. The work item is assigned to 'NT' (Nguyễn Chí Tài) and has a status of 'DONE'.

24127545 - Nguyễn Ngọc Thiên - 25%

- Task Evidence:

My Software Team

Summary | **Backlog** | Board | Code | Timeline | Docs | Forms

Search backlog | Filter

S0 - Proposal 9 Jun - 14 Jun (5 work items) | Complete sprint

Item	Assignee	Status	Due Date
SCRUM-1 [Tài] Template 0 - các mục p...	NT	DONE	Jun 14
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	NT	DONE	Jun 13
SCRUM-22 [Thiên] Template 0 - các mục...	NT	DONE	Jun 14
SCRUM-23 [Quân] Template 0 - các mục...	QA	DONE	Jun 14
SCRUM-24 [Tiến] Template 0 - các mục...	TQ	IN PROGRESS	Jun 14

Jira work item details:

- Assignee: NT Nguyễn Ngọc Thiên
- Parent: None
- Due date: Jun 14, 2026
- Labels: None
- Team: None
- Start date: Jun 09, 2026
- Sprint: S0 - Proposal
- Story point estimate: None
- Reporter: QA Quân Phạm Trần Anh

My Software Team

Summary | **Backlog** | Board | Code | Timeline | Docs | Forms

Search backlog | Filter

S0 - Proposal 9 Jun - 14 Jun (5 work items) | Complete sprint

Item	Assignee	Status	Due Date
SCRUM-1 [Tài] Template 0 - các mục p...	NT	DONE	Jun 14
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	NT	DONE	Jun 13
SCRUM-22 [Thiên] Template 0 - các mục...	NT	DONE	Jun 14
SCRUM-23 [Quân] Template 0 - các mục...	QA	DONE	Jun 14
SCRUM-24 [Tiến] Template 0 - các mục...	TQ	IN PROGRESS	Jun 14

Jira work item details:

Viết Mục 3.1.1 - Features (user story mapping)

Done | Agents | Done | Improve Subtask

Description
Add a description...

Linked work items
Add linked work item

Details:

- Assignee: NT Nguyễn Ngọc Thiên
- Labels: None
- Parent: SCRUM-22 [Thiên] Template 0 - các mục
- Sprint: S0 - Proposal
- Story point estimate: None
- Reporter: QA Quân Phạm Trần Anh

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' tab with a search bar and filters. The main area lists several sprints: 'S0 - Proposal' (9 Jun - 14 Jun, 5 work items), 'S1 - Requirements' (15 Jun - 28 Jun, 4 work items), 'S2 - Design' (29 Jun - 5 Jul, 4 work items), 'S3 - Testing' (6 Jul - 12 Jul, 5 work items), and 'S4 - Defense Prep' (13 Jul - 20 Jul, 4 work items). The 'S0 - Proposal' sprint is expanded, showing a list of work items: 'SCRUM-1 [Tải] Template 0 - các mục p...' (DONE, Jun 14), 'SCRUM-2 Chốt đề tài & phạm vi MeoMa...' (DONE, Jun 13), 'SCRUM-22 [Thiên] Template 0 - các mục...' (DONE, Jun 14), 'SCRUM-23 [Quân] Template 0 - các mục...' (DONE, Jun 14), and 'SCRUM-24 [Tiến] Template 0 - các mục...' (IN PROGRESS, Jun 14). The right sidebar shows the 'Jira work item' detail view for 'Viết Mục 3.1.2 - Architecture + vẽ sơ đồ'. The 'Details' section shows the Assignee as 'NT Nguyễn Ngọc Thiên', Labels as 'None', Parent as 'SCRUM-22 [Thiên] Template 0 - các mục', Sprint as 'S0 - Proposal', Story point estimate as 'None', and Reporter as 'QA Quân Phạm Trần Anh'.

The screenshot displays the Jira interface for the 'My Software Team' space. The left sidebar shows the 'Backlog' tab with a search bar and filters. The main area lists several sprints: 'S0 - Proposal' (9 Jun - 14 Jun, 5 work items), 'S1 - Requirements' (15 Jun - 28 Jun, 4 work items), 'S2 - Design' (29 Jun - 5 Jul, 4 work items), 'S3 - Testing' (6 Jul - 12 Jul, 5 work items), and 'S4 - Defense Prep' (13 Jul - 20 Jul, 4 work items). The 'S0 - Proposal' sprint is expanded, showing a list of work items: 'SCRUM-1 [Tải] Template 0 - các mục p...' (DONE, Jun 14), 'SCRUM-2 Chốt đề tài & phạm vi MeoMa...' (DONE, Jun 13), 'SCRUM-22 [Thiên] Template 0 - các mục...' (DONE, Jun 14), 'SCRUM-23 [Quân] Template 0 - các mục...' (DONE, Jun 14), and 'SCRUM-24 [Tiến] Template 0 - các mục...' (IN PROGRESS, Jun 14). The right sidebar shows the 'Jira work item' detail view for 'Self-training: Gemini function calling'. The 'Details' section shows the Assignee as 'NT Nguyễn Ngọc Thiên', Labels as 'None', Parent as 'SCRUM-22 [Thiên] Template 0 - các mục', Sprint as 'S0 - Proposal', Story point estimate as 'None', and Reporter as 'QA Quân Phạm Trần Anh'.

The screenshot shows the Jira interface for 'My Software Team'. The left pane displays the 'Backlog' with a list of work items under the 'S0 - Proposal' sprint. The right pane shows the details of a selected work item titled 'Review Mục 2, Mục 7'.

Backlog Items:

Item	Assignee	Status	Due Date
SCRUM-1 [Tải] Template 0 - các mục p...	NT	DONE	Jun 14
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	NT	DONE	Jun 13
SCRUM-22 [Thiên] Template 0 - các mục...	NT	DONE	Jun 14
SCRUM-23 [Quân] Template 0 - các mục...	QA	DONE	Jun 14
SCRUM-24 [Tiến] Template 0 - các mục...	TQ	IN PROGRESS	Jun 14

Work Item Details: Review Mục 2, Mục 7

- Assignee:** NT Nguyễn Ngọc Thiên
- Labels:** None
- Parent:** SCRUM-22 [Thiên] Template 0 - các mục
- Sprint:** S0 - Proposal
- Story point estimate:** None
- Reporter:** QA Quân Phạm Trần Anh

The screenshot shows the Jira interface for 'My Software Team'. The left pane displays the 'Backlog' with a list of work items under the 'S0 - Proposal' sprint. The right pane shows the details of a selected work item titled 'Quay phần thuyết trình của mình'.

Backlog Items:

Item	Assignee	Status	Due Date
SCRUM-1 [Tải] Template 0 - các mục p...	NT	DONE	Jun 14
SCRUM-2 Chốt đề tài & phạm vi MeoMa...	NT	DONE	Jun 13
SCRUM-22 [Thiên] Template 0 - các mục...	NT	DONE	Jun 14
SCRUM-23 [Quân] Template 0 - các mục...	QA	DONE	Jun 14
SCRUM-24 [Tiến] Template 0 - các mục...	TQ	IN PROGRESS	Jun 14

Work Item Details: Quay phần thuyết trình của mình

- Assignee:** NT Nguyễn Ngọc Thiên
- Labels:** None
- Parent:** SCRUM-22 [Thiên] Template 0 - các mục
- Sprint:** S0 - Proposal
- Story point estimate:** None
- Reporter:** QA Quân Phạm Trần Anh

2

Preliminary Problem Statement

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127250 – Phan Quang Tiến

Reviewed by: 24127226 – Phạm Trần Anh Quân

Context and Motivation

Electronic mail remains a critical communication channel for academic and professional purposes. However, users frequently encounter **information overload**, receiving dozens to hundreds of emails daily, including work notifications, academic updates, advertisements, and spam.

Managing this volume of data introduces major pain points:

- **Time-Consuming Management:** Users must spend substantial time manually opening, reading, and filtering individual emails to extract vital information or perform routine maintenance.
- **Inefficient Legacy Systems:** Traditional email filtering and searching mechanisms are rigid and unintelligent, operating primarily on static keyword matching or hard-coded rules. They are incapable of understanding semantic context, user intent, or providing core summaries.

To address these challenges, this project proposes **MeoArc**—an intelligent email application that integrates a Large Language Model (LLM) agent as its core orchestration engine. MeoArc acts as a virtual assistant, enabling users to manage, summarize, categorize, and interact with their mailboxes through natural language commands. Beyond its built-in assistant, MeoArc is designed as an Email Intelligence Platform. The platform exposes email-management capabilities through REST APIs and an MCP server, enabling external AI agents to securely access and perform email-related operations under user authorization. This allows MeoArc to support not only human users but also AI-driven workflows and future agent-based applications.

Operating Environment:

The software functions within a cross-platform, distributed environment consisting of the following layers:

- **Client Side:** The application operates smoothly on modern web browsers supporting HTML5, CSS3, and standard ECMAScript (such as Google Chrome, Microsoft Edge, or Mozilla Firefox).
- **AI Gateway & Server Side:** Backend services are orchestrated via **FastAPI**, serving as a high-performance **AI Gateway** to manage asynchronous request routing, prompt formatting, and secure communication. The entire backend system is containerized using Docker to guarantee seamless deployment, isolated dependencies, and uniform execution. The server can be hosted locally (Localhost) for testing or deployed on Cloud Servers to handle live traffic.
- **Third-Party Integrations:** The backend establishes secure connections with the Google Gmail API to synchronize email threads and metadata. Concurrently, the **FastAPI AI Gateway** connects with the Google Gemini API to leverage state-of-the-art generative AI capabilities for intent recognition and function calling execution

Design and Implementation Constraints

The system will be developed as a **web-based application** to ensure accessibility across different platforms without requiring software installation.

React with **TypeScript** is selected for frontend development because it supports component-based architecture, code maintainability, and efficient user interface rendering.

FastAPI is chosen for backend and AI Gateway development due to its high performance, asynchronous processing capabilities, and seamless integration with Python-based AI libraries.

PostgreSQL is utilized as the primary database because it provides reliable relational data management, strong consistency, and support for complex queries.

Redis is employed for caching and temporary session storage to improve application responsiveness and reduce unnecessary processing overhead.

Docker is adopted to provide a consistent development and deployment environment across team members' machines.

The system must use **OAuth 2.0** when accessing Gmail services and must comply with Google's authentication and security requirements.

All communications involving user data and external services must be performed through secure HTTPS connections.

The project documentation, reports, and deliverables must conform to the standards and templates specified by the course requirements.

3

Proposed Solution

3.1 Software

3.1.1. Features

Written by: 24127545 – Nguyễn Ngọc Thiên

Edited by: 24127226 – Phạm Trần Anh Quân

Reviewed by: 24127529 – Nguyễn Chí Tài

Demand	Feature
As a student living in the age of rapidly advancing AI, I prefer interacting with AI through natural conversation rather than clicking buttons or navigating complex menus.	Fully AI-driven chat interface powered by natural language processing (NLP).
As a user, when the agent searches, drafts, or processes multiple emails, I want to see the results visually (email lists, previews, etc.) instead of reading a long block of text.	Dynamic canvas interface displaying email cards, draft previews, bulk-action selection panels, and a real-time timeline of the agent's workflow.
As an HR manager, I wake up every day to dozens of unread emails that require attention.	Email summarization and task extraction.
As an office worker constantly dealing with deadlines, I frequently need to write emails to different audiences. Each email requires a different	AI-powered email drafting, reply generation, and automatic sending.

tone, level of formality, and wording to ensure professionalism and respect.	
As the owner of a small business, I receive a large number of spam emails every day, such as stock investment and real estate promotions, which bury important customer emails.	Bulk email management (delete, mark as read, label) and contextual email search/filtering.
As a user, I want to remain in control of everything, especially sensitive data such as my emails.	Human-in-the-loop workflow: Plan → Execute or request clarification → User approval → Automated execution → Completion notification.
As a user, I want the application to support my preferred language and allow customization to suit my device and preferences.	Settings management (language, theme, and personalization options).
As a user, I do not want the application to request unnecessary permissions, nor do I want to risk losing access to my account.	OAuth authentication (Google/Microsoft), secure data handling, and granular email permission management with support for multiple connected accounts.
As a user, I find it inconvenient to log in every time I access the application.	Session management with JWT (access + refresh token)
As a user, I want to be able to revisit previous conversations whenever needed	Conversation history management.
As a user, I do not want to wait more than three minutes for the application to complete a task or provide a response.	High-performance processing and response system with asynchronous execution, background jobs, and caching.
As a developer, I want to protect my application from Distributed Denial-of-Service (DDoS) attacks and ensure it cannot be abused to attack others.	Rate limiting and abuse prevention mechanisms.
As a developer, I need visibility into system failures and issues so that bugs can be identified and resolved efficiently.	Comprehensive logging and monitoring system.
As a developer, I need to maintain the system within the planned operational budget.	Token usage optimization through prompt engineering and cost-efficient AI orchestration.
As a power user with my own AI assistant (e.g., Claude, a custom agent), I want to control my	Agent-facing interface via MCP (Model Context Protocol) server, exposing

mailbox through my own agent instead of opening the web application.	MeoArc's email tools (search, summarize, draft, send, bulk actions) with structured schemas so external agents can operate the application.
As a busy professional, I want the AI assistant to perform common multi-step email workflows automatically so that I can focus on high-priority tasks.	Reusable AI Skills for email workflows, including: • Email Triage Skill: classify unread emails, identify priorities, summarize important messages, and suggest recommended actions. • Daily Digest Skill: collect emails received within a selected period, group them by category, and generate a concise summary report. • Meeting Preparation Skill: analyze related email threads, extract action items, deadlines, and discussion points, then generate a meeting brief.
As a user, I want to know how much of my mailbox history the assistant is allowed to look at under my current plan, and to be able to move to a higher plan when I need more.	Subscription tiers — Free, Pro and Pro Max — each with its own AI processing scope (90, 180 and 365 days of mailbox history respectively), a persistent indicator of the current scope beside the chat input, and a plan comparison screen.
As a first-time visitor, I want to understand what MeoArc does and what it will be allowed to touch in my mailbox before I sign in with my email account.	Public landing page presenting the value proposition, the assistant's capabilities, the seven email categories, the subscription plans, and the sign-in entry point.

3.1.2. Software Architecture

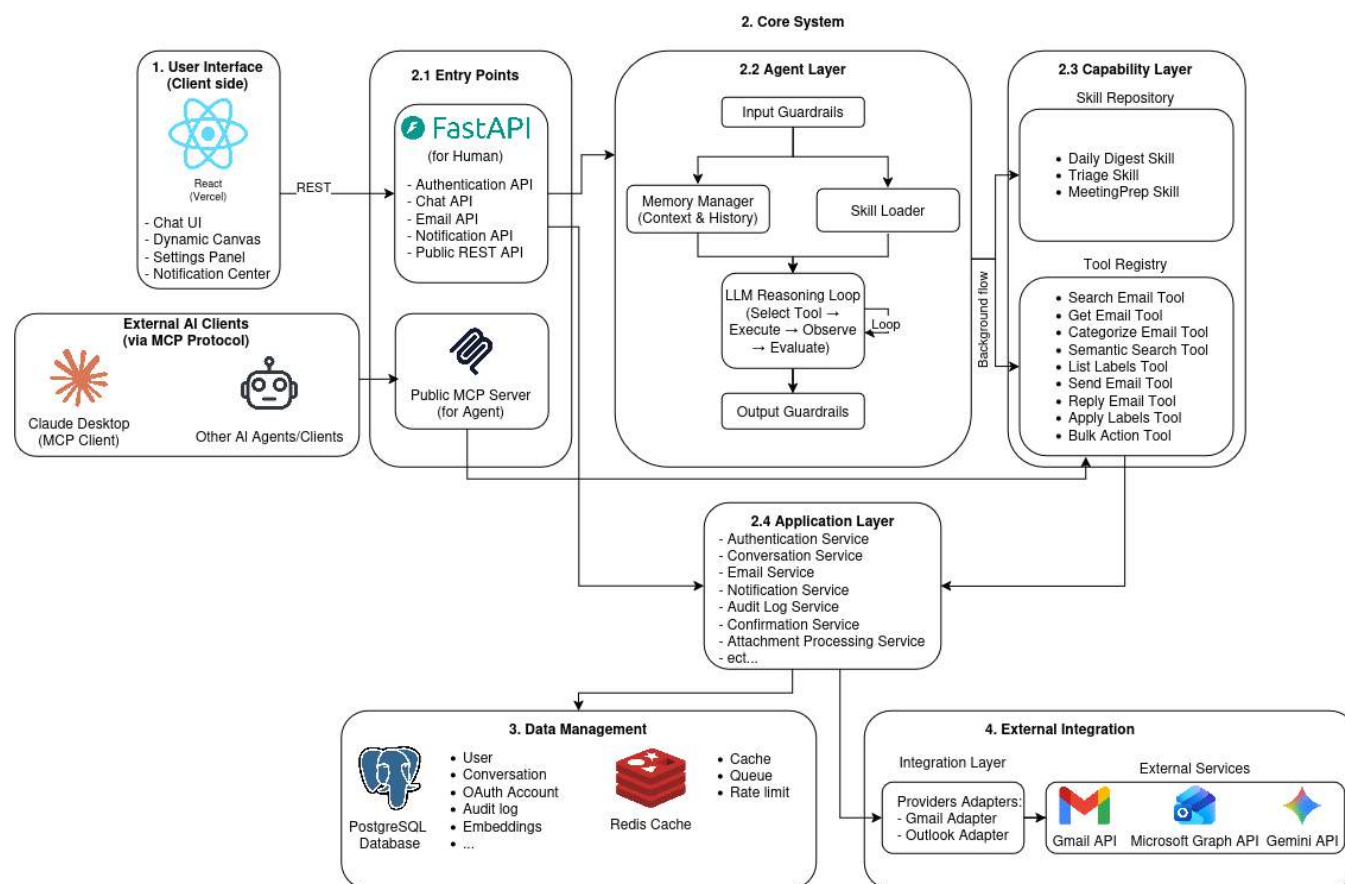
Please include the ID and full name of the student who authored, reviewed, or updated this section.

Written by: 24127545 - Nguyễn Ngọc Thiên

Edited by: 24127529 - Nguyễn Chí Tài

Reviewed by: 24127226 - Phạm Trần Anh Quân

The student team should present the overall architecture of the software system that the team will build to fulfill the list of functionalities outlined in section 3.1.1.



*External agents (e.g., Claude Desktop) connect through the Public MCP Server to operate MeoArc's tools without opening the web app.

3.2 Hardware

Written by: 24127250 - Phan Quang Tiến

Edited by: 24127545 - Nguyễn Ngọc Thiên

Reviewed by: 24127226 – Phạm Trần Anh Quân

The student team should describe the hardware and equipment requirements necessary for the software to operate.

1. Client-Side Requirements

Users are required to have a personal computer or laptop with a stable internet connection (min: 10Mbps) to access the MeoArc web application through a modern browser.

The recommended minimum specifications are: a dual-core CPU (2.0GHz or above), 4GB RAM and any operating system supporting HTML5-compatible browsers such as Google Chrome, Microsoft Edge or Mozilla Firefox.

2. Development Environment Requirements

Each team member's development machine must be capable of running Docker Desktop alongside the full development stack (FastAPI backend, PostgreSQL, Redis and the React frontend concurrently).

The recommended minimum specifications are: a quad-core CPU, 8GB RAM (16 recommended), 20GB of available disk space and a stable Internet connection for accessing the Gmail API and Gemini API during development and testing.

3. Server-Side Requirements

As the project operated entirely on free-tier cloud platforms (Render for backend, Vercel for frontend) and local Docker containers during development, no dedicated physical server hardware is required. The Render free-tier instance provides sufficient compute resources for handling the expected low-to-moderate traffic during the academic evaluation period.

4

Development Plan

Written by: 24127529 – Nguyễn Chí Tài

Edited by: 24127545 – Nguyễn Ngọc Thiên

Reviewed by: 24127226 – Phạm Trần Anh Quân

4.1 Requirements Analysis

Duration: Week 1

The objective of this phase is to identify system requirements, define the project scope, and establish the foundation for subsequent design and development activities.

Activities:

- **Email Structure Analysis (BE & AI):** Analyze Gmail API email payloads to identify available metadata and content fields. The results will support PostgreSQL database design and determine the information accessible to the LLM.
- **User Flow Analysis (FE):** Analyze the end-to-end user journey, including authentication, email management, and AI interactions. The resulting workflow will guide UI/UX design and Figma prototyping.
- **Natural Language Command Analysis (AI & BE):** Identify common user commands and map them into structured intents and parameters. This analysis will define the backend functions required for AI-assisted email management.
- **MVP Scope Definition (All Members):** Establish clear project boundaries and limitations to ensure feasibility within the six-week development schedule. Features beyond the MVP scope will be excluded to prevent scope creep.

Deliverables:

- Software Requirement Specification (SRS)
- User Flow Diagram
- Functional and Non-Functional Requirements
- MVP Scope Definition Document

4.2 Software Design

Duration: Week 2

The objective of this phase is to transform the identified requirements into a detailed software design that guides the implementation process.

Activities:

- Design the overall system architecture, including Frontend, FastAPI AI Gateway, Backend Services, PostgreSQL, Redis, Gmail API, and Gemini API.
- Design the PostgreSQL database schema and Redis data structures based on the analyzed Gmail data.
- Design the Tool Registry and MCP server interface, including tool schemas, tool discovery mechanisms, authentication for agent connections, and standardized request/response formats.
- Define API specifications, including request/response formats and communication between system components.
- Design the AI workflow, including intent recognition, function calling, and response generation.
- Design the Google OAuth 2.0 authentication flow and security mechanisms.
- Create UI/UX wireframes and Figma prototypes for key application screens.
- Prepare the initial test plan and testing strategy.

Deliverables:

- Software Architecture Diagram
- Database Schema Design
- API Specification Document
- Tool Registry Specification
- AI Workflow Design
- AI Skills Specification
- OAuth 2.0 Flow Diagram
- Figma Prototype
- Test Plan

4.3 Implementation

Duration: Weeks 3 – 4

The objective of this phase is to develop the software components based on the approved design specifications.

Activities:

- Implement the FastAPI backend and integrate PostgreSQL and Redis.
- Implement the MCP server exposing email tools to external agents.
- Implement the Tool Registry responsible for managing available email operations, tool metadata, and agent-accessible capabilities.
- Develop Google OAuth 2.0 authentication and Gmail API integration.
- Implement email management functionalities, including synchronization, search, and categorization.
- Integrate Gemini API and implement AI features such as email summarization and natural language interaction.
- Develop frontend interfaces based on the Figma design.
- Configure Docker environment and version control workflows.
- Conduct unit testing throughout development to ensure component reliability.

Deliverables:

- Functional Frontend Application
- Backend Services and REST APIs
- Database and Redis Integration
- Gmail API Integration
- AI Assistant Features
- Source Code Repository (/src)
- Docker Configuration
- MCP Server

4.4 Testing

Duration: Week 5

The objective of this phase is to verify that the implemented system satisfies the specified requirements and operates reliably under expected usage conditions.

Activities:

- Execute unit testing for frontend, backend, and AI-related components.

- Perform integration testing between the Frontend, Backend, Database, Gmail API, and Gemini API.
- Validate authentication, email management, and AI assistant functionalities.
- Identify, document, and fix defects discovered during testing.
- Conduct user acceptance testing (UAT) to ensure the system meets user expectations.
- Test agent-facing channel with an external MCP client.

Deliverables:

- Test Cases and Test Reports
- Bug Tracking Report
- Integration Testing Results
- User Acceptance Testing Report
- Updated and Stable Software Version

4.5 Deployment and Maintenance

Duration: Week 6

The objective of this phase is to deploy the completed system and ensure its stable operation.

Activities:

- Prepare the production environment using Docker Compose.
- Deploy the Frontend, Backend, PostgreSQL, and Redis services.
- Configure environment variables and external service integrations, including Gmail API and Gemini API.
- Perform final system verification after deployment.
- Monitor application logs and system performance.
- Fix critical issues discovered during deployment and testing.
- Prepare project documentation and user guides.

Deliverables:

- Deployed Application
- Production Docker Configuration
- Deployment Guide
- User Manual
- Final Project Documentation

5

Human Resources & Costing Plan

Written by: 24127529 - Nguyễn Chí Tài

Edited by: 24127545 - Nguyễn Ngọc Thiên

Reviewed by: 24127226 - Phạm Trần Anh Quân

5.1 Human Resources Allocation

The project team consists of four members, each responsible for a primary area while collaborating throughout the software development lifecycle.

Member	Role	Responsibilities
Nguyễn Chí Tài	Backend Lead	Design and implement backend services using FastAPI, integrate PostgreSQL and Redis, develop REST APIs, and manage Gmail API integration.
Nguyễn Ngọc Thiên	AI Lead	Design AI workflows, implement Gemini API integration, develop prompt engineering strategies, and support function-calling mechanisms.
Phạm Trần Anh Quân	Frontend Lead	Design and implement the user interface, create Figma prototypes, develop frontend components, and integrate frontend services with backend APIs.
Phan Quang Tiến	QA Engineer (Tester)	Prepare test plans, execute testing activities, track defects, validate system requirements, and support deployment verification.

The responsibilities of team members across development phases are summarized below:

Phase	Participants	Responsibilities
Requirements Analysis	All Members	Analyze requirements, define MVP scope, identify user stories and system constraints.
Software Design	All Members	Design architecture, database schema, APIs, AI workflow, and UI/UX prototype.
Implementation	Backend Lead, AI Lead, Frontend Lead	Develop backend services, AI functionalities, and frontend interfaces.
Testing	QA Engineer, All Members	Execute testing activities, report defects, and verify fixes.
Deployment & Maintenance	All Members	Deploy the system, monitor performance, and prepare final documentation.

5.2 Costing Plan

As this project is developed for academic purposes, the team prioritizes free-tier services and open-source technologies to minimize operational costs.

Resource Category	Service Provider	Allocation & Tier Configuration	Actual Cost (VND)
Container Platform	Docker Desktop / WSL2 (development)	Running FastAPI, PostgreSQL, and Redis in isolated local containers	0 VND
LLM Service	Gemini API	Free Tier (15 RPM, 1,500 RPD)	0 VND
Relational Database	PostgreSQL + pgvector → PostgreSQL + pgvector (Neon)	Free Tier	0 VND

Cache & Session Store	Redis Cloud	Free Tier	0 VND
Cloud Hosting (BE + FE)	Render (backend), Vercel (frontend)	Free Tier	0 VND
External Mail Service	Google Gmail API	Access through developer Google accounts	0 VND
UI/UX & Project Management	Figma, Jira, Overleaf	Free / Student Tiers	0 VND
Source Code Management	GitHub	Free Team Repository	0 VND

The project is expected to operate entirely within free-tier service limits during the development and evaluation period.

6

Tools setup

Written by: 24127250 – Phan Quang Tiến

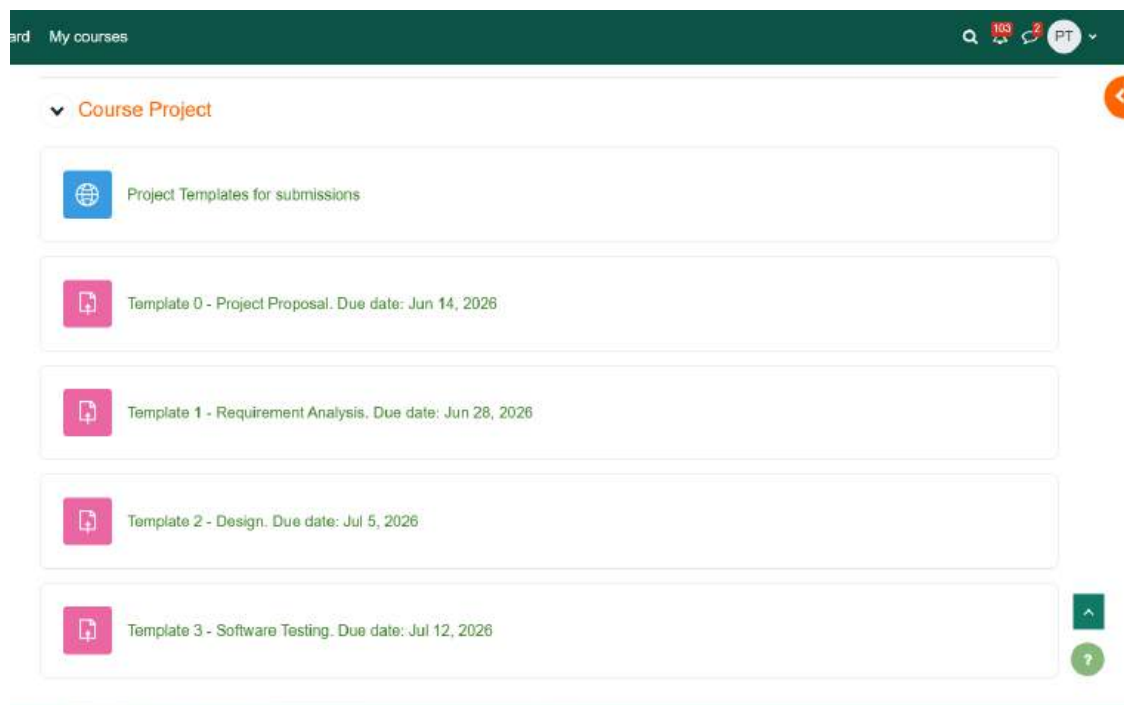
Edited by: 24127545 – Nguyễn Ngọc Thiên

Reviewed by: 24127226 – Phạm Trần Anh Quân

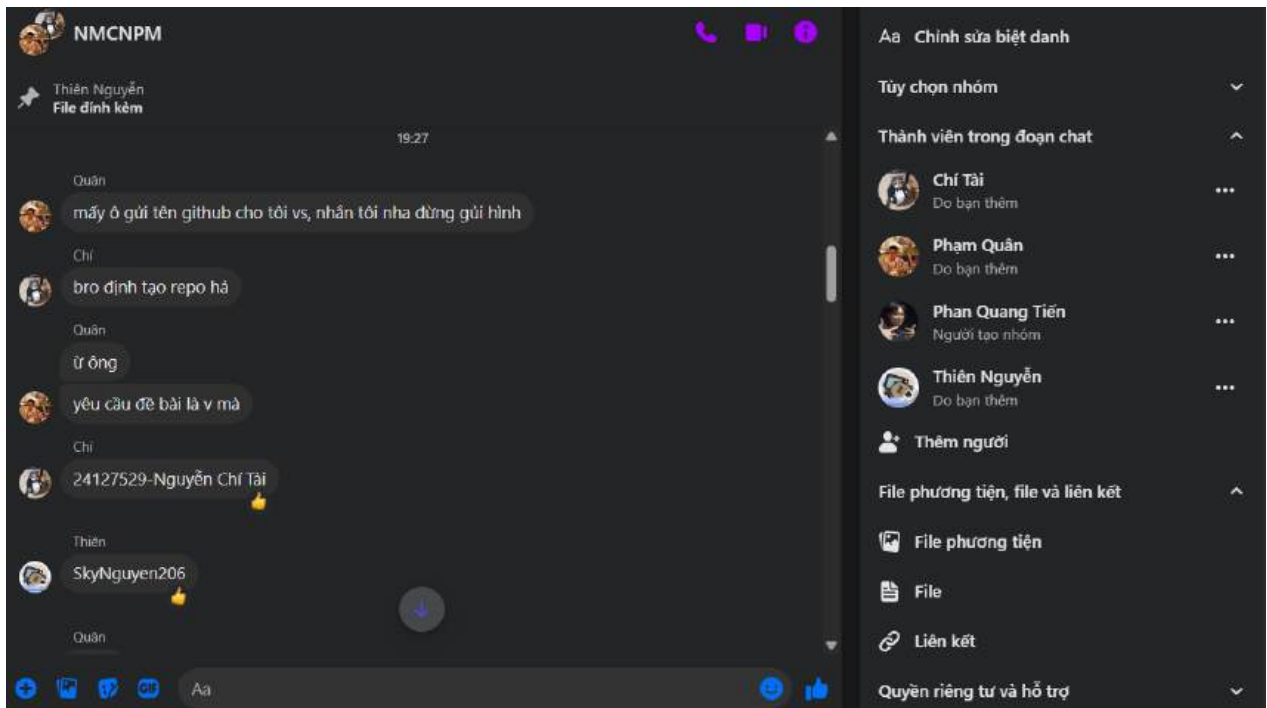
One of the main goals of this project assignment is to help you practice teamwork for professional software development. We'll assess and evaluate your effectiveness in team collaboration. Everyone is required to work closely with others in the group to deliver results, and we'll use tools to monitor and assess each group's teamwork performance.

You are required to use the following tools:

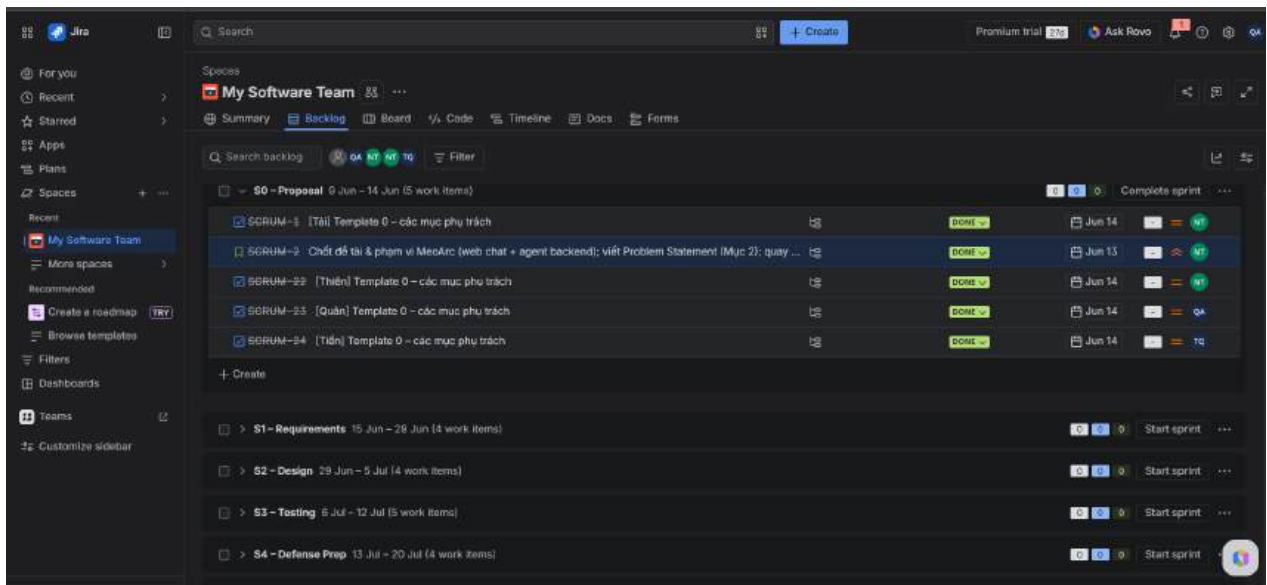
- **Moodle:** Used for posting and submitting assignments.

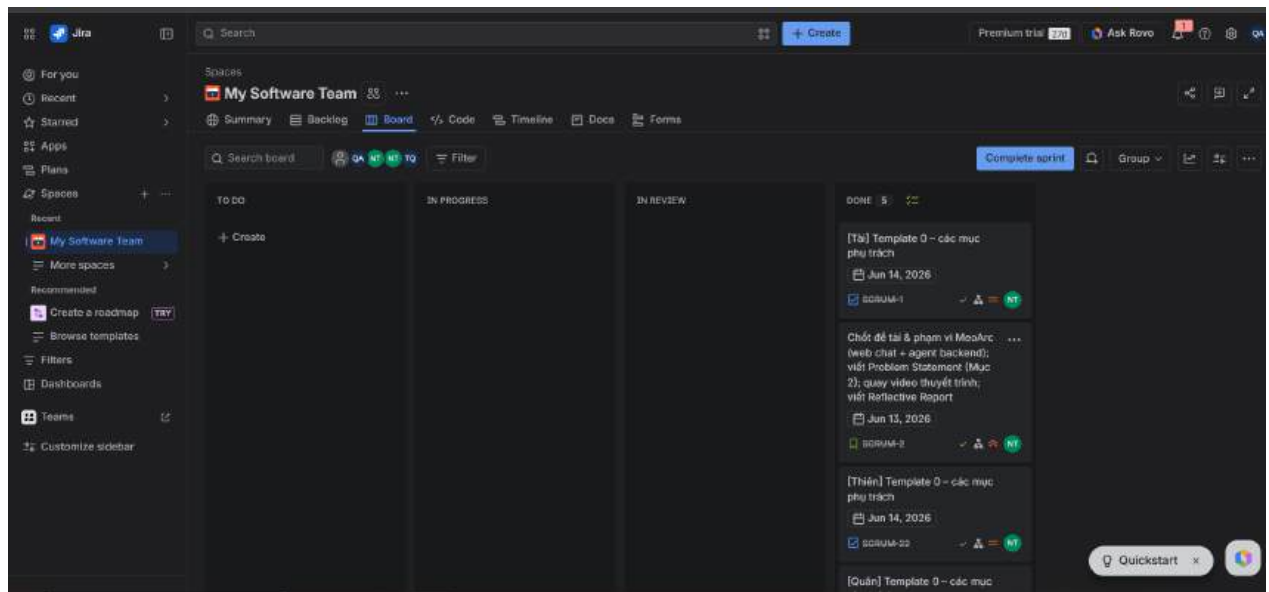


- **Messenger Group:** Used for discussions and interactions among group members for each project.

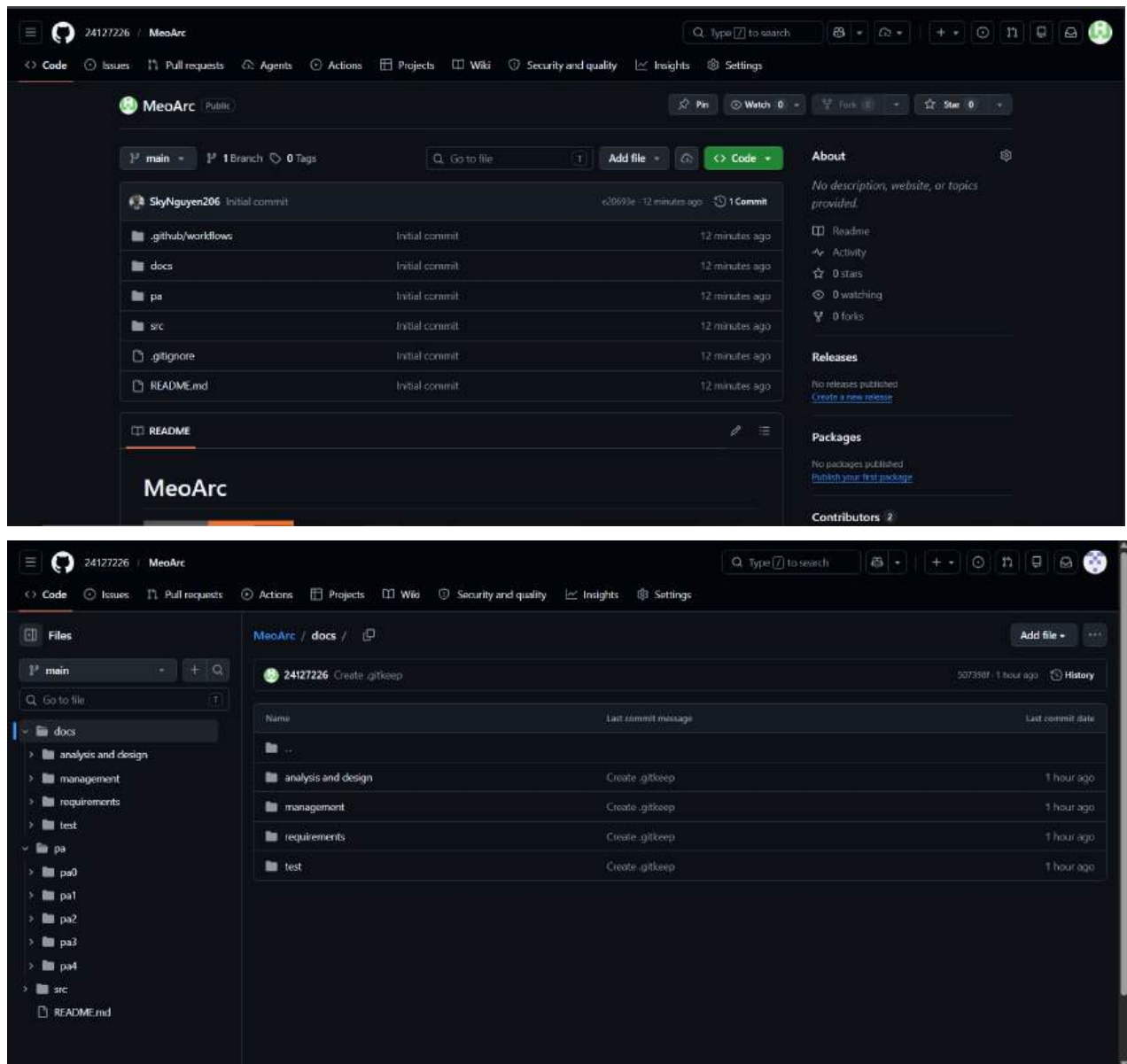


- JIRA: Each group will have one board. Tasks are organized into sprints.



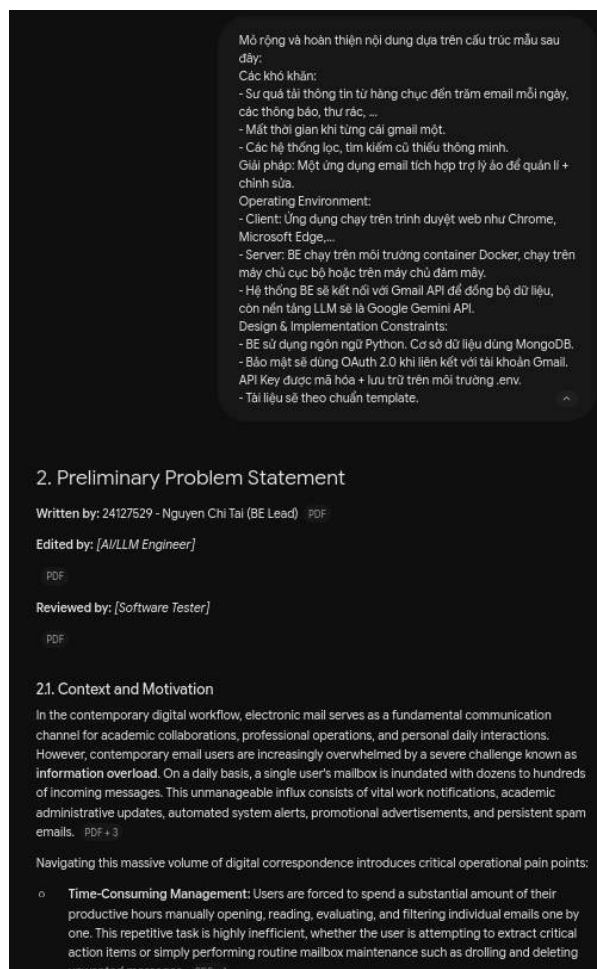


- **GitHub:** Used to store source code and documentation. Each group will belong to a team on the repository and have the following folders:
 - /src: Used to store source code.
 - /docs: Used to store documentation, which has these subfolders:
 - management: Storing planning documents, reports (weekly report, project status report, etc.).
 - requirements: Storing all requirements, including vision documents and use cases.
 - analysis and design: Storing all analysis and design related documents, including software architecture documents, UML models, and UI designs.
 - test: Storing all test documents such as test plans, test cases, and test reports.
 - /pa: Including subfolders to store submissions. Each subfolder contains one PA submission.
- TAs will help set up these tools during the first few lab sessions.



Important note: the effective use of tools to support teamwork is crucial for project success. Thus, using tools effectively is an important criterion for grading. Everyone is required to use the tools chosen by the team regularly.

7 AI Usage Declaration



Gemini. Gemini 3.5 Flash, Google, gemini.google.com, accessed 14:00 on June 9, 2026, prompt: “Mô rộng và hoàn thiện nội dung dựa trên cấu trúc mẫu sau đây: Các khó khăn... Giải pháp...

Operating Environment... Design & Implementation Constraints...”, used for the preliminary

problem statement section in Chapter 2; AI provided rough content, student rewrote in academic

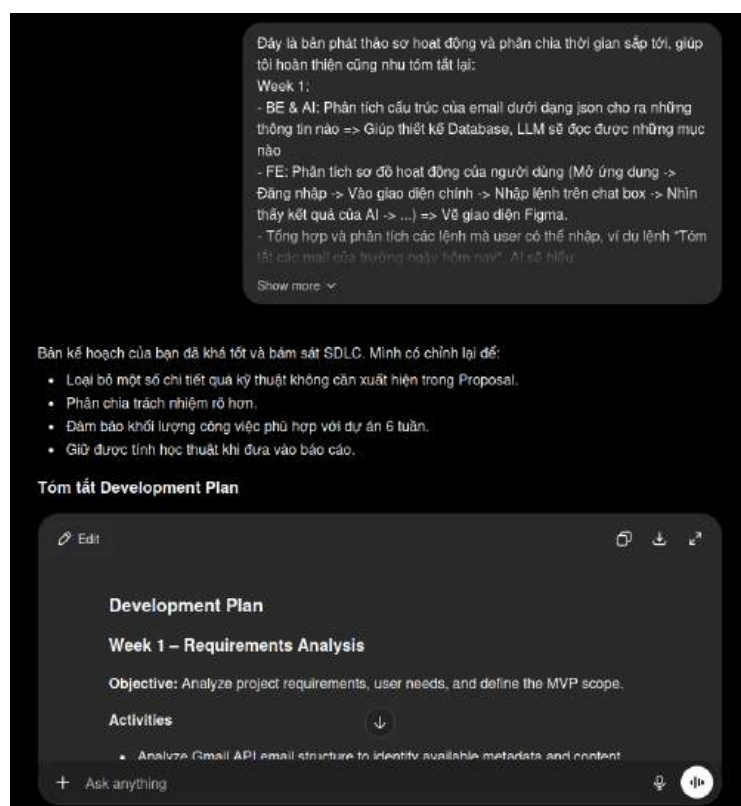
language, expanded operational pain points, and added official HCMUS template standards.

ChatGPT. GPT-4o, chatgpt.com, accessed 15:35 on June 9, 2026, prompt: “Ngoài các chức năng nghiệp vụ mà tôi đã đưa cho bạn trong file idea.md, bên phía chức năng hệ thống / phi hệ thống có gì không, như Đăng nhập OAuth, quản lý phiên bằng JWT hay bảo mật chẳng hạn”, this is used to analyze other system and user requirements for section 3.1.1; AI provided me with 12 additional system/non-system functions, some of which repeat what I've already outlined, and some are designed for deployment scenarios. <[gpt-chat-history](#)>

ChatGPT. GPT-4o, chatgpt.com, accessed 16:10 on June 9, 2026, prompt: “nếu chúng tôi chỉ là sinh viên, không đc hỗ trợ cho phí phát triển ứng dụng này, thì việc deploy và maintance nên làm như thế nào, liệu là dùng Render + Vercel để deploy, Ollama/gemini-api để có thể maintance chức năng AI chính vì đang là free không”, used for choose a deployment tool that can calculate costs appropriate for the current project. <[gpt-chat-history-deploy & maintenance](#)>

ChatGPT. GPT-4o, chatgpt.com, accessed 17:11 on June 9, 2026, prompt: “nhận xét tôi vẽ overall architecture of the software system như vậy được chưa, còn điểm nào sai, hay tôi bị nhầm lẫn chỗ nào”, used for Software architecture drawing support, part 3.1.2.

<[gpt-chat-history-software-architecture](#)>



ChatGPT. GPT-4o, chatgpt.com, accessed 19:13 on June 9, 2026, prompt: “Đây là bản phát thảo sơ hoạt động và phân chia thời gian sắp tới, giúp tôi hoàn thiện cũng như tóm tắt lại: Week 1 ... Week 2 ... Week 3-4 ... Week 5 ... Week 6”, used for present activities according to milestones. in Chapter 4; AI provided rough content, students rewrote in academic language, expanded operational pain points, and added official HCMUS template standards.

ChatGPT. GPT-4o, chatgpt.com, accessed 16:44 on June 10, 2026, prompt: “Với yêu cầu sau đây, bạn nghĩ tôi đã vẽ đúng software architecture chưa, có chỉnh sửa gì không: The student team should present the overall architecture of the software system that the team will build to fulfill the list of functionalities outlined in section 3.1.1 đây là phần 3.1.1 ...”, used for Software architecture drawing support, part 3.1.2. <[gpt-chat-history-architecture](#)>

ChatGPT. GPT-4o, chatgpt.com, accessed 15:54 on June 10, 2026, prompt: “hãy giúp tôi phiên dịch các đoạn sau thành tiếng anh, chỉnh sửa lại cho chuyên nghiệp và đúng ngữ pháp, trả về dưới dạng table có 2 cột là demand và feature ...”, used for translating Section 3.1.1 to English

<[gpt-chat-translate](#)>

8

Presentation

Written by: 24127226 – Phạm Trần Anh Quân

Edited by: 24127545 – Nguyễn Ngọc Thiên

Reviewed by: 24127529 – Nguyễn Chí Tài

Teams are required to record presentation videos of their work using the provided template.

Every team member must participate in the presentation, specifically covering the sections their contributed to the project.

Videos must not exceed 30 minutes.

Please upload your videos to Youtube (set to either Unlisted or Public).

Provide the Youtube links the the field below.

[GROUP 7 _ PROJECT PROPOSAL](#)

9

Reflective Report

Which sections of this template are most helpful for implementation, and which are unnecessary? Provide specific examples to support your reasoning.

Written by: 24127250 - Phan Quang Tiến

Edited by: 24127529 - Nguyễn Chí Tài

Reviewed by: 24127545 - Nguyễn Ngọc Thiên

- Sections are helpful for implementation:
 - **Section 2 – Preliminary Problem Statement:** is one of the most valuable sections for implementation. It forced the team to clearly define the problem scope, operating environment and technical constraints from the very beginning. For MeoArc specifically, establishing OAuth 2.0, FastAPI and PostgreSQL as hard constraints early on prevented architectural changes later in development.
 - **Section 3.1.1 – Features:** was also highly useful because framing requirements as user stories pushed the team to think from the user's perspective rather than a purely technical one, making MVP prioritization much easier.
 - **Section 4 – Development Plan:** is equally critical. By breaking the project into weekly phases with explicit deliverables, the team always had a clear picture of what needed to be done and when, which significantly reduced ambiguity during execution.
- Sections are unnecessary:
 - **Section 1 – Member Contribution Assessment:** feels premature at the proposal stage, since actual work has not yet begun and contribution percentages cannot be accurately estimated this early; however, we recognize it establishes accountability and a shared sense of ownership among team members from the outset.
 - **Section 8 – Presentation:** has little direct impact on software implementation. It serves a reporting purpose but does not contribute to the actual building of the

system. Nevertheless, it remains valuable as it requires the team to consolidate the entire project into a coherent narrative and to communicate technical decisions clearly—a skill that is essential in professional software development.